Permutation & Combination

1. Fundamental Counting Principle

Q1.A person has 5 shirts, 4 trousers and 3 pairs of shoes. In how many different ways can the person select one shirt, one trouser and one pair of shoes?

A) 12
B) 20
C) 60
D) 80

Q2.

A password consists of 2 letters followed by 3 digits. Letters can be repeated, and digits can also be repeated. How many different passwords are possible?

A) 67,600

B) 676,000

C) 6,760

D) 260,000

Q3. There are 4 roads from Hyderabad to Vijayawada and 3 roads from Vijayawada to Chennai. How many different routes are possible from Hyderabad to Chennai through Vijayawada?

A) 7
B) 12
C) 24
D) 16

Q4. How many three-digit numbers can be formed using the digits 1, 2, 3, 4 and 5 without repetition?

A) 25

B) 60

C) 100

D) 125

Q5. A test contains 6 questions. Each question has 4 options, and exactly one option must be selected for each question. How many different answer patterns are possible?

A) 24

B) 256

C) 1,024

D) 4,096

2. Factorials

Q1. Find the value of:

7!/5!

A) 35
B) 42
C) 49
D) 56

Q2.

If: $$\frac{n!}{(n-2)!} = 30$$

Find the value of n.

A) 5

B) 6

C) 7

D) 8

Q3. Evaluate the Expression

Find:

$$\frac{8!}{6! \times 2!}$$

A) 14

B) 28

C) 56

D) 112

Q4. What is the value of:

$$\frac{5! + 4!}{3!}$$

A) 20
B) 24
C) 28
D) 32

Q5. Find the value of:

$$\frac{10!}{8!} + \frac{6!}{5!}$$

A) 86
B) 92
C) 96
D) 102

3. Permutations

Q1. In how many ways can 5 students stand in a straight line?

A) 25

B) 60

C) 100

D) 120

Q2. From 8 students, a president and a secretary must be selected. One student cannot hold both positions. How many ways can these positions be filled?

A) 28
B) 56
C) 64
D) 72

Q3. How many different arrangements can be made using all the letters of the word CAT?

A) 3

B) 6

C) 9

D) 12

Q4. In how many ways can 6 different books be arranged on a shelf if 2 particular books must always be together?

A) 120
B) 240
C) 360
D) 720

When 2 particular objects must always be together:

$$(n-1)! \times 2!$$

Q5. How many four-digit numbers can be formed using 1, 2, 3, 4, 5 and 6 without repetition?

A) 120

B) 240

C) 360

D) 720

4. Combinations

Q1. A teacher wants to select 3 students from a group of 7 students. How many different selections are possible?

A) 21
B) 28
C) 35
D) 42

Q2. From 10 employees, a committee of 4 members is formed. How many different committees are possible?

A) 120

B) 180

C) 210

D) 240

Q3. A group contains 5 boys and 4 girls. In how many ways can a team of 3 students be selected with exactly 2 boys and 1 girl?

A) 20
B) 30
C) 40
D) 60

Total ways $= {}^{a}C_{b} \times {}^{c}C_{d}$

Q4. A student must answer 5 questions out of 8 questions. How many different selections are possible?

A) 40

B) 48

C) 56

D) 64

Q5.A team of 4 members must be selected from 6 men and 5 women. If the team must contain exactly 2 men and 2 women, how many teams are possible?

A) 100
B) 120
C) 150
D) 180

5. Arrangement Problems

Q1. In how many ways can 6 people sit in a row if A and B must always sit together?

A) 120

B) 240

C) 360

D) 720

Arrangement	Formula
Straight row	n!
Circular table	(n-1)!
2 people together in a row	(n-1)! x 2!

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Q2. How many arrangements of the word APPLE are possible if the two P's must always be together?

A) 12

B) 24

C) 36

D) 48

Q3. In how many ways can 5 friends sit around a circular table?

A) 24

B) 60

C) 120

D) 240

Q4. **In how many ways can 5 boys and 3 girls stand in a row if all 3 girls must stand together?**

A) 720

B) 2,880

C) 3,600

D) 4,320

Q5. How many arrangements can be made using all letters of the word GARDEN if the vowels must always occupy the first two positions?

A) 120

B) 48

C) 360

D) 64

6. Selection Problems

Q1. A team of 5 members must be selected from 8 players. How many different teams can be formed?

A) 40
B) 48
C) 56
D) 64

Q2. A committee of 3 members is formed from 5 men and 4 women. How many committees contain at least one woman?

A) 64
B) 74
C) 84
D) 94

Q3. A student has 7 different books and wants to select 4 books. If one particular book must always be selected, how many selections are possible?

A) 10

B) 15

C) 20

D) 35

Q4. A group contains 6 boys and 5 girls. How many committees of 4 members can be formed with exactly 2 girls?

A) 100

B) 120

C) 150

D) 180

Q5. A committee of 4 members is formed from 7 men and 5 women. How many committees can be formed with at least 2 men?

A) 420

B) 315

C) 350

D) 385